

Summer Math Adventure

W2
D1

Week 2 · Day 1 | Fractions: Meaning & Equivalent Fractions

Name: _____

Date: _____

FRACTIONS

Fraction = part of a whole. $\frac{\text{numerator}}{\text{denominator}}$ → numerator = how many parts you have, denominator = total equal parts.

Equivalent fractions: multiply OR divide numerator AND denominator by the same number.

$$\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$$

Part A – Shade & Label

Shade the correct fraction of each bar. Then write an equivalent fraction by doubling.

1. Shade $\frac{3}{4}$ — then find an equivalent fraction with denominator 8.

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Equivalent: $\frac{3}{4} = \frac{\boxed{}}{8}$

2. Shade $\frac{2}{5}$ — then find an equivalent fraction with denominator 15.

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Equivalent: $\frac{2}{5} = \frac{\boxed{}}{15}$

Part B – Find the Missing Number

3. $\frac{3}{5} = \frac{\boxed{}}{20}$

4. $\frac{4}{6} = \frac{2}{\boxed{}}$

5. $\frac{5}{8} = \frac{\boxed{}}{40}$

6. $\frac{18}{24} = \frac{3}{\boxed{}}$

Part C – Simplest Form

Divide numerator and denominator by their GCF (Greatest Common Factor).

7. $\frac{8}{12} = \frac{\boxed{}}{\boxed{}}$

8. $\frac{15}{20} = \frac{\boxed{}}{\boxed{}}$

9. $\frac{24}{36} = \frac{\boxed{}}{\boxed{}}$

10. $\frac{30}{45} = \frac{\boxed{}}{\boxed{}}$

Part D – Compare Fractions

Write $<$, $>$, or $=$. First convert to a common denominator.

11. $\frac{3}{4} \boxed{} \frac{5}{8}$

Common denom: _____

12. $\frac{2}{3} \boxed{} \frac{3}{5}$

Common denom: _____

$$13. \frac{7}{8} \square + \frac{11}{12}$$

Common denom: _____

$$14. \frac{4}{6} \square - \frac{6}{9}$$

Common denom: _____

Summer Math Adventure

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D2

Week 2 · Day 2 | Adding & Subtracting Fractions with Unlike Denominators

Name: _____

Date: _____

★ Challenge – Fraction Order

FRACTIONS: ADD & SUBTRACT

Order these fractions from least to greatest. Show your work using a common

To add or subtract fractions with different denominators:

Step 1: Find the **LCD** (Least Common Denominator)

Step 2: Convert each fraction to an equivalent fraction using the LCD

Step 3: Add or subtract the numerators. Keep the denominator.

Step 4: Simplify if needed.

Example: $\frac{1}{3} + \frac{1}{4} \rightarrow \text{LCD} = 12 \rightarrow \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$

Part A – Find the LCD First

1. Denominators: 4 and 6 LCD = $\square$

2. Denominators: 3 and 8 LCD = $\square$

3. Denominators: 5 and 6 LCD = $\square$

4. Denominators: 4 and 9 LCD = $\square$

Part B – Add the Fractions

Show the LCD step. Simplify your answer.

5. $\frac{1}{2} + \frac{1}{5} =$

LCD: $\square$

$$\frac{\square}{\square} + \frac{\square}{\square} =$$

6. $\frac{2}{3} + \frac{1}{4} =$

LCD: $\square$

$$\frac{\square}{\square} + \frac{\square}{\square} =$$

7. $\frac{3}{4} + \frac{2}{5} =$

LCD: $\square$

=

8. $\frac{5}{6} + \frac{3}{8} =$

LCD: $\square$

=

Part C – Subtract the Fractions

9. $\frac{3}{4} - \frac{1}{3} =$

10. $\frac{5}{6} - \frac{1}{4} =$

$$11. \frac{7}{8} - \frac{2}{3} =$$

$$12. 1 - \frac{3}{7} =$$

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D3

Summer Math Adventure

Week 2: Day 3 | Mixed Numbers: Converting, Adding & Subtracting

Name: _____

Date: _____

MIXED NUMBERS

Mixed Number: whole number + fraction (e.g., $3\frac{1}{2}$)

Convert to improper fraction: multiply whole $\times$ denominator, add numerator, keep denominator.

$$3\frac{1}{2} \rightarrow (3 \times 2) + 1 = 7 \rightarrow \frac{7}{2}$$

Convert to mixed number: divide. Quotient = whole, remainder = new numerator.

$$\frac{11}{4} \rightarrow 11 \div 4 = 2 \text{ R } 3 \rightarrow 2\frac{3}{4}$$

Part A – Convert Between Forms

Mixed $\rightarrow$ Improper:

$$1. 2\frac{3}{4} = \frac{\boxed{}}{\boxed{}}$$

14. A recipe needs $\frac{3}{4}$ cup flour. You only have $\frac{1}{3}$ cup. ~~How much more flour do you need?~~

$$2. 5\frac{2}{3} = \frac{\boxed{}}{\boxed{}}$$

Answer:

$$3. 4\frac{5}{6} = \frac{\boxed{}}{\boxed{}}$$

Improper $\rightarrow$ Mixed:

$$4. \frac{17}{5} =$$

$$5. \frac{29}{4} =$$

$$6. \frac{41}{6} =$$

Part B – Challenge Fraction Chain

Fill in the blank fractions to make the chain equal 1. Add whole + add fractions together. If fraction ≥ 1 , if fraction ≥ 1 .

$$7. 3\frac{1}{2} + 2\frac{1}{4} =$$

$$\frac{1}{3} + \frac{1}{4} + \frac{\boxed{}}{\boxed{}} = 1$$

$$8. 4\frac{2}{3} + 1\frac{3}{4} =$$

Show your work:

$$9. 2\frac{5}{6} + 3\frac{3}{4} =$$

$$10. 6\frac{7}{8} + 2\frac{3}{4} =$$

Part C – Subtract Mixed Numbers

Summer Math Adventure Grade 5 Week 2 Day 3

Mixed Numbers

If the fraction being subtracted is **BIGGER**, you must borrow 1 from the whole number.

$$11. 5\frac{3}{4} - 2\frac{1}{4} =$$

$$12. 6\frac{1}{3} - 2\frac{2}{3} = (\text{need to borrow!})$$

Summer Math Adventure

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D4

Week 2 · Day 4 | Multiplying Fractions & Fractions of Whole Numbers

Name: _____

Date: _____

MULTIPLYING FRACTIONS (need to borrow!)

$$14. 10 - 4\frac{2}{5} =$$

Multiply fractions: top × top, bottom × bottom.

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$

Fraction of a whole number: multiply the whole number by the fraction.

$$\frac{3}{4} \text{ of } 24 = \frac{3}{4} \times 24 = \frac{3 \times 24}{4} = \frac{72}{4} = 18$$

Tip: Cross-cancel before multiplying to keep numbers small!

★ Challenge – Pizza Party!

Part A – Multiply Fractions

At a party, Zach ate $2\frac{3}{8}$ pizzas, Ana ate $1\frac{3}{4}$ pizzas, and Dom ate $1\frac{1}{2}$ pizzas. They started with 6 pizzas total.

$$1. \frac{1}{2} \times \frac{3}{5} =$$

$$2. \frac{3}{4} \times \frac{2}{9} =$$

$$3. \frac{5}{6} \times \frac{3}{10} =$$

A. How much pizza did all three eat together?

$$4. \frac{4}{7} \times \frac{7}{8} =$$

$$5. \frac{2}{3} \times \frac{6}{7} =$$

$$6. \frac{5}{8} \times \frac{4}{5} =$$

B. How much pizza was left over?

Part B – Fraction of a Whole Number

$$7. \frac{3}{4} \text{ of } 32 =$$

$$8. \frac{2}{5} \text{ of } 45 =$$

$$9. \frac{5}{6} \text{ of } 54 =$$

$$10. \frac{7}{8} \text{ of } 64 =$$

Part C – Word Problems

11. A hiking trail is $4\frac{1}{2}$ miles. Sofia walked $\frac{2}{3}$ of the trail. How far did she walk?

Answer:

12. A class of 30 students: $\frac{3}{5}$ chose soccer, $\frac{1}{6}$ chose swimming, and the rest chose basketball. How many chose each sport?

Summer Math Adventure

Soccer:

W2
D5

Week 2 · Day 5 | Fractions on Number Lines & Cumulative Review

Swimming: _____

Date: _____

FRACTIONS: REVIEW

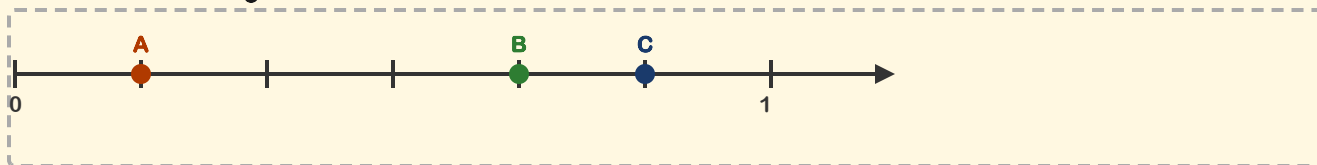
Number line tip: The space between 0 and 1 is divided into equal parts equal to the denominator. Count "jumps" equal to the numerator.

Part A – Number Line Fractions

★ Challenge – Area with Fractions

Label the marked points on each number line.

1. A garden plot is $\frac{1}{6}$ yd wide and $\frac{5}{6}$ yd long. What is its area? (Area = length $\times$ width)
Each section = $\frac{1}{4}$. Label A, B, C.



A = _____ B = _____ C = _____

2. Plot these fractions on the number line: $\frac{1}{4}$, $\frac{3}{4}$, $\frac{1}{2}$
square yards



Part B – Mixed Review Sprint

3. $\frac{3}{8} + \frac{1}{4} =$

4. $\frac{5}{6} - \frac{1}{3} =$

5. $\frac{3}{5} \times \frac{2}{3} =$

6. $3\frac{1}{2} + 2\frac{3}{4} =$

7. $5\frac{1}{3} - 1\frac{2}{3} =$

8. $\frac{2}{3}$ of 27 =

Part C – Cumulative: Number Sense + Fractions

9. A bucket holds $3\frac{3}{4}$ gallons. You fill it $\frac{2}{3}$ full. How many gallons are in the bucket?

Answer:

10. A school raised \$48,360. They spent $\frac{3}{8}$ on supplies. How much was spent on supplies?

Summer Math Booklet · Grade 4 → 5 · Week 2 Day 5

Fractions Review

(Hint: find $\frac{3}{8}$ of 48,360)

Week 2 Quiz – Fractions

W2
QUIZ

Answer: \$

Topics: Equivalence · Comparing · Add/Subtract · Mixed Numbers · Multiply

Name: _____

Date: _____

Score: _____ / 20

★ Challenge – Fraction Reasoning

Without calculating exactly, decide which sum is larger. Explain your reasoning.

A: $\frac{1}{8} + \frac{5}{6}$ vs B: $\frac{3}{4} + \frac{11}{12}$

1. Fill in the blank. Why? = $\frac{\square}{28}$

2. Simplify: $\frac{20}{28} = \frac{\square}{\square}$

Check by calculating:

3. Compare: $\frac{5}{8}$ $\square$ $\frac{3}{5}$

4. Order least to greatest: $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{8}$

5. (1 pt) Convert $6\frac{3}{5}$ to an improper fraction.

Section 2 – Add & Subtract (7 pts)

6. $\frac{1}{3} + \frac{1}{6} =$

7. $\frac{5}{6} - \frac{3}{8} =$

8. $4\frac{3}{4} + 2\frac{2}{3} =$

9. $7\frac{1}{4} - 3\frac{3}{4} =$

10. (3 pts) Lena has $5\frac{1}{2}$ feet of ribbon. She uses $2\frac{3}{8}$ feet for one gift and $1\frac{3}{4}$ feet for another.

How much ribbon remains?

Section 3 – Multiply (5 pts + bonus)

11. $\frac{3}{5} \times \frac{5}{9} =$

12. $\frac{3}{4} \times 36 =$

13. (3 pts) A rectangular room is $9\frac{1}{2}$ ft long and 8 ft wide. What is the area?

Answer:

sq ft